Isospin

$\binom{p}{n}$ isospin doublet $\quad |p\rangle = |1/2, 1/2\rangle \quad |n\rangle = |1/2, -1/2\rangle$

deuterium $\quad |pn\rangle \overset{?}{=} |0,0\rangle \quad$ verify the singlet hypothesis

$|d\rangle = |pn \text{ bound state}\rangle$

scattering exp. to produce deuterium.

$p \longrightarrow$ target $\qquad \dfrac{dN_r}{dt} = \sigma \cdot \dfrac{dN_p}{dt} \cdot n_b \cdot d$

n_b ↓ increase density of target

${}^{9}_{4}Be$, ${}^{63}_{29}Cu$: typical targets

$p + Be \longrightarrow \begin{cases} p+p \longrightarrow & p+p \\ p+n \longrightarrow & p+n \end{cases}$ } elastic

$\Rightarrow$ look inelastic for deuterium production

$\sigma \propto |M|^2 \rho(E)$

(1) $p + p \longrightarrow d + x^+$

	p	p	d	x^+
Q	1	1	1	1
B	1	1	2	0
I	$\frac{1}{2}$	$\frac{1}{2}$	0	1
I_3	$+\frac{1}{2}$	$+\frac{1}{2}$	0	+1

$\Rightarrow x^+ \equiv \pi^+$

$$|p+p\rangle = |1/2, 1/2\rangle |1/2, 1/2\rangle = |1, +1\rangle \quad (I,\ I_3)$$

Final State: $\quad d + \pi^+ = |0,0\rangle |1,+1\rangle$

$$0 \otimes 1 = 1$$

$$|d+\pi^+\rangle = |1,1\rangle$$

if strong interaction conserves isospin

$\langle f | H_I | i \rangle \qquad |i\rangle, |f\rangle$ must have the same isospin.

$$\sigma \propto |M|^2 \rho(E)$$

$$M = \langle f | H_I | i \rangle = \langle d + \pi^+ | H_{strong} | p+p \rangle$$

from isospin: $\langle d+\pi^+ | H_I | p+p\rangle = \langle 1,1 | H_I | 1,1\rangle = \mathcal{M}_a$

(2) $p + n \longrightarrow d + X^0$

	p	n	d	X^0
Q	+1	0	+1	0
B	+1	+1	+2	0
I	1/2	1/2	0	1
I_3	+1/2	-1/2	0	$\emptyset$

$\} \Rightarrow X^0 \equiv \pi^0$

$\frac{1}{2} \otimes \frac{1}{2} = 0, 1$

to be verified experimentally

initial state $|i\rangle$

$|p+n\rangle$: $I_3 = \emptyset$, $I = 0, 1$ $\qquad \frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$

$$|p+n\rangle = \frac{1}{\sqrt{2}}\left(|0,0\rangle + |1,0\rangle\right)$$

Final state $|d+\pi^0\rangle$

$I_3 = \emptyset$ $\qquad |0,0\rangle |1,0\rangle = |1,0\rangle$

$0 \otimes 1 = 1$

$$|i\rangle = \frac{1}{\sqrt{2}}\left(|0,0\rangle + |1,0\rangle\right) \xrightarrow[\text{strong}]{H_I} |f\rangle = |1,0\rangle$$

$$\mathcal{M}_b = \langle f | H_I | i\rangle = \langle 1,0 | H_I \frac{1}{\sqrt{2}}\left(\cancel{|0,0\rangle} + |1,0\rangle\right)$$

if H_I(strong) conserves isospin I

$$\mathcal{M}_b = \frac{1}{\sqrt{2}} \langle 1,0 | H_I | 1,0\rangle$$

$\sigma \propto$ # events.

$$\sigma_a(p+p\to d+\pi^+) \propto |\mathcal{M}_a|^2 \rho(p+p\to d+\pi^+)$$

$$\sigma_b(p+n\to d+\pi^0) \propto |\mathcal{M}_b|^2 \rho(p+n\to d+\pi^0)$$

$$\frac{\sigma_a}{\sigma_b} = \underbrace{\frac{\#\, p+p\to d+\pi^+}{\#\, p+n\to d+\pi^0}}_{\text{experimental meas.}} = \frac{|\mathcal{M}_a|^2}{|\mathcal{M}_b|^2} \frac{\rho(d\pi^+)}{\rho(d\pi^0)}$$

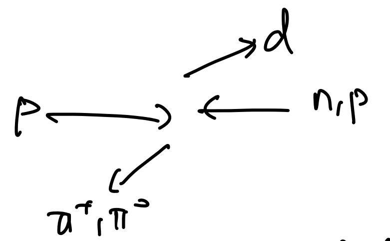
$$\frac{|M_a|^2}{|M_b|^2} = \frac{|\langle 1,1|H_I|1,1\rangle|^2}{|\frac{1}{\sqrt{2}}\langle 1,0|H_I|1,0\rangle|^2} = (\sqrt{2})^2 = 2$$

$\langle I=1 | H_I | I=1\rangle$
cance between a, b

$$\frac{\rho(p+p\to d^* + \pi^+)}{\rho(p+n\to d+\pi^0)}$$

Same becau of protons with same energy E

$\frac{\Delta m_{pn}}{m_p} \simeq 1/1000$ $\qquad$ $\frac{\Delta\pi}{m_\pi} \simeq \frac{5}{140} \sim 5/100$



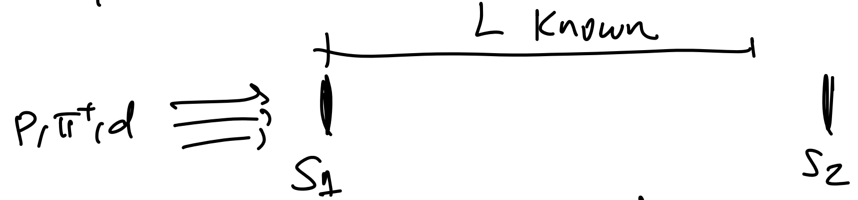
$\Rightarrow$ within 5% $\qquad \frac{\rho(d\pi^+)}{\rho(d\pi^0)} \simeq 1$

Experimentally: $d+\pi^+$, $d+\pi^0$

π^+: charged track observed w/ scintillators
trackers

Counting π^+ $\Rightarrow$ counting # $p+p\to d+\pi^+$

You have to discriminate $p+p \not\to p+p$

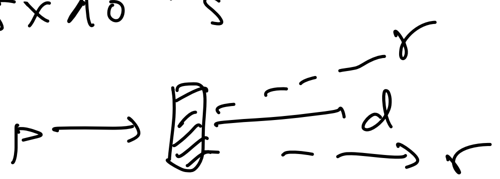


: scintillators.

measure $\Delta t = \frac{L}{v}$

$p+n \to d+\pi^0$ $\qquad$ $\pi^0 \to \gamma\gamma$ EM decay

$\tau_{\pi^0} = 8.5\times 10^{-17}$ s



observe γ
lack of other tracks.

$\Rightarrow$ exp. discriminate $d\pi^+$ from $d\pi^0$

$$\frac{\#\ \pi^+ d}{\#\ \pi^0 d} = \frac{\sigma_a}{\sigma_b} \simeq 2 \quad \text{experimentally}$$

$\Rightarrow$ confirms 1/ isospin observed

2/ $|d\rangle = |0,0\rangle$

Another experiment

$(p/n) + \underset{\text{target}}{(p/n)} \longrightarrow$

anti-protons:

$p + p \longrightarrow p + p + p + \bar{p}$

$p + p + \pi^+ + \pi^-$

Reduces phase space for $p + p \rightarrow p + p + X$

$\pi^+/\pi^- + p/n \longrightarrow$

$|I=1, I_3 = \pm 1\rangle \qquad |1/2, +1/2, -1/2\rangle$

$p + p \longrightarrow p + p + \pi^0$

$p + p + \pi^+ + \pi^-$